Differential Amplifier with Current-Mirror Bias

Lab report

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Date	October 27th 2026		

1 Introduction

The current mirror biased the differential pair at a tail current of 1.87 mA, 37% above the 1.37 mA design value. The Early effect and an estimated 6 mV V_{BE} mismatch between the two discrete mirror transistors account for the difference. The measured DC operating point, the pre-lab mirror simulations and the output swing limits agree with analysis. The small-signal quantities could not be measured on the bench: the base signal in step 2 was below the oscilloscope noise floor, the only output capture was taken with the pair overdriven, and the common-mode output was not a response to the input. They are reported instead from hand analysis and from an LTspice simulation matched to the measured bias, which agree within 3%: $A_d = -77.4$ V/V, $R_{\mathrm{id}} = 8.6$ kΩ, and a CMRR of 63.4 dB single-ended and 94.6 dB differential.

2 Objectives

- Use a BJT current mirror to supply the bias (tail) current of a differential pair.
- Measure the pair's DC operating point, input resistance, differential gain, common-mode gain and CMRR, and compare each with analysis and simulation.

3 Pre-lab: current mirror

3.1 Hand analysis

With $V_{\mathrm{BE}} \approx 0.7$ V, the reference current set by the 6.8 kΩ resistor is

$$i_{\mathrm{ref}} = \frac{0 - 0.7 - (-10)}{6.8\ \mathrm{k\Omega}} = 1.37\ \mathrm{mA}. \tag{1}$$

Q_2 stays active while its collector-emitter voltage exceeds $V_{\mathrm{CE,sat}} = 0.2$ V:

$$V_{\mathrm{CE2}} = 10 - i_{\mathrm{copy}} R_x \geq 0.2\ \mathrm{V} \quad \Rightarrow \quad 0 \leq R_x \leq \frac{9.8\ \mathrm{V}}{i_{\mathrm{copy}}} = 7.16\ \mathrm{k\Omega}. \tag{2}$$

Above this bound Q_2 saturates and $i_{\mathrm{copy}} \approx 9.8\ \mathrm{V}\ / R_x$, for example 0.98 mA at 10 kΩ.

3.2 Simulation

The mirror was simulated in Multisim (Figure 1, Figure 2) and, to separate the non-ideal effects, in LTspice with three versions of the 2N3394 model (Table 1, Figure 3; netlist in Appendix B). V_{AF} was raised to 2×10^6 V in every case, so the Early effect plays no role.

Table 1: Current-mirror results. Effective $\beta = 2/(i_{\text{ref}}/i_{\text{copy}} - 1)$. In all three LTspice cases $i_{\text{copy}} = 0.991$ mA at $R_x = 10$ kΩ.

Model	i_{ref}	i_{copy} **at** $R_x = 1$ **kΩ**	$i_{\text{copy}}/i_{\text{ref}}$	**Effective** β	R_x **at** $V_{\text{CE2}} = 0.2$ **V**
Hand analysis (ideal)	1.37 mA	1.37 mA	1	∞	7.16 kΩ
(1) 2N3394 as given, $\beta_F = 121.3$	1.380 mA	1.337 mA	0.969	62	7.36 kΩ
(2) β_F raised to 10^6	1.380 mA	1.359 mA	0.985	130	7.24 kΩ
(3) β_F raised and $I_{\text{SE}} = 0$	1.380 mA	1.380 mA	1.000	$\approx \infty$	7.13 kΩ

The ideal model (3) reproduces the hand analysis: $i_{\text{copy}} = i_{\text{ref}}$, and the saturation bound agrees within 0.5% (7.13 against 7.16 kΩ). Its i_{ref} is 1.380 rather than 1.368 mA only because the model's V_{BE} is 0.62 V rather than 0.7 V. Raising β_F alone is not enough. The model's base-emitter leakage term ($I_{\text{SE}} = 1.22 \times 10^{-13}$ A, $N_E = 1.305$) adds about 10 µA of base current at 1.4 mA, which limits the effective β to about 130 in case (2) and to about 62 with the default β_F in case (1). The two Multisim results correspond to these cases: the operating point of Figure 1 (1.359 mA) to case (2), and the sweep of Figure 2 (flat at 1.34 mA) to case (1), so the sweep was run on the unedited model. In every case Q_2 leaves the active region between 7.1 and 7.4 kΩ.

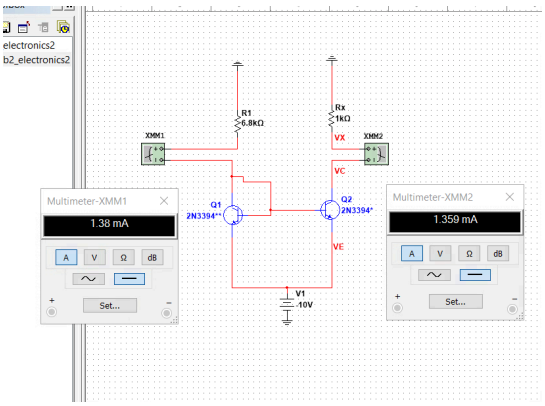


Figure 1: Multisim mirror at $R_x = 1$ kΩ with β_F raised: $i_{\text{ref}} = 1.38$ mA, $i_{\text{copy}} = 1.359$ mA.

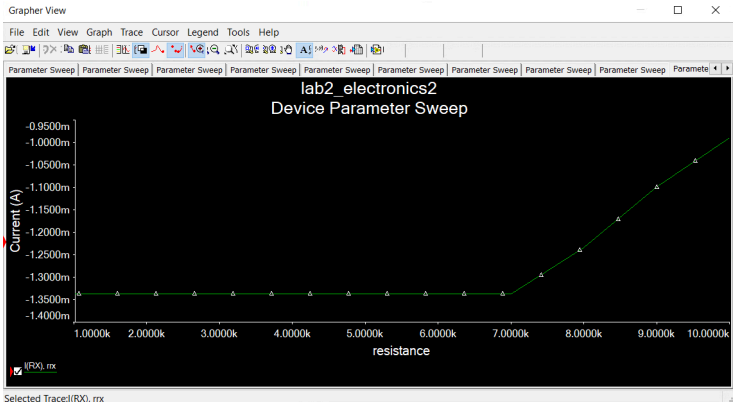


Figure 2: Multisim sweep of R_x (unedited model): i_{copy} is flat at 1.34 mA up to about 7 kΩ, then falls as Q_2 saturates. The negative sign is the reference direction.

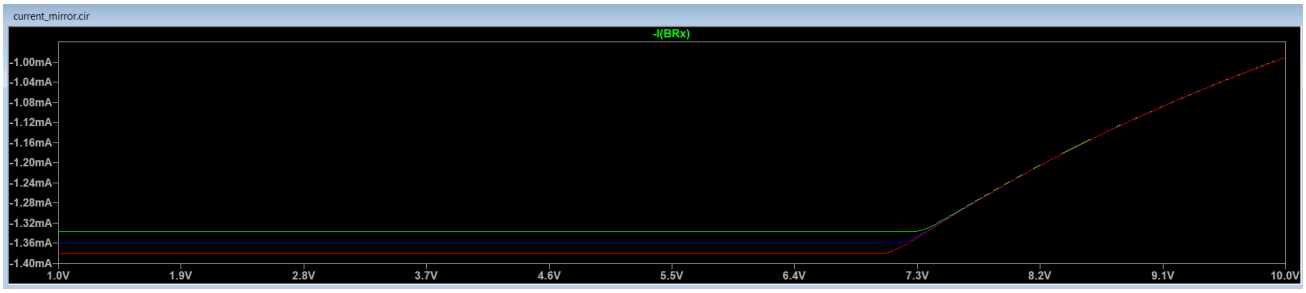


Figure 3: LTspice sweep of R_x for the three models of Table 1 (green: case 1, blue: case 2, red: case 3). The horizontal axis V_{rx} is R_x in kΩ; the current is plotted negative to match the Multisim sign convention.

4 Circuit and setup

The mirror ($R_K = 6.8$ kΩ, emitters at −10 V) sets the tail current of the Q_1, Q_2 pair (2N3904, $R_C = 2.2$ kΩ, $R_B = 2$ MΩ to +10 V, $C_1 = 10$ µF), as shown in Figure 4. In differential mode the base of Q_2 was grounded at A; in common mode it was moved to B. R_S was a decade resistance box. For step 2 it was

set to 9 kΩ with an 820 kΩ resistor in series, so $R_S = 829$ kΩ. Component and supply values were not measured, so nominal values are used throughout.

Simulation of the amplifier. The circuit of Figure 4 was simulated in LTspice (Appendix B). To compare like with like, the simulation was matched to the measured bias. The pair transistors use $\beta_F = 137$ and 129, which give the measured effective β of 156 and 147 at $V_{CE} \approx 11$ V with $V_A = 74$ V. The mirror's output transistor has 1.28 times the area of the reference transistor, which represents the 6 mV V_{BE} mismatch estimated in Section 5.1. The DC agreement in Table 3 is therefore by construction; the small-signal results are the prediction being tested. Gains were computed with an AC analysis at 5 kHz, and amplitudes, waveforms and output limits with a transient analysis.

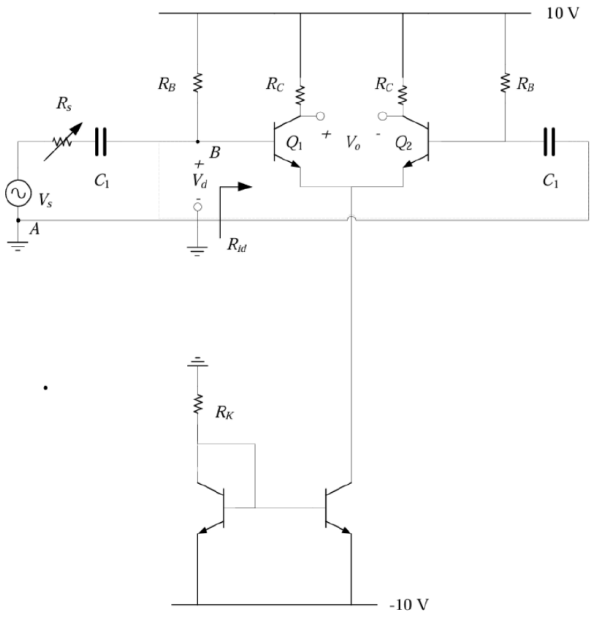


Figure 4: Differential amplifier with current-mirror tail (Lab Manual 2, Figure 7).

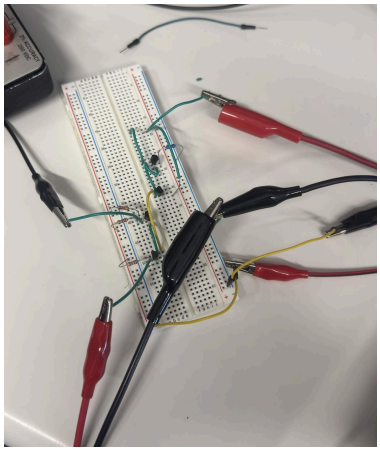


Figure 5: Breadboard build with supply and probe leads.

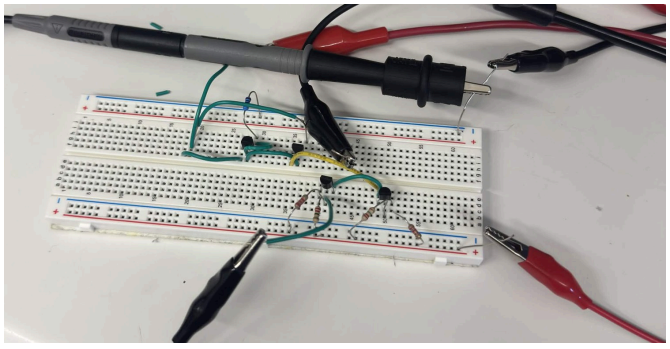


Figure 6: Breadboard with the oscilloscope probe connected for the AC measurements.

5 Results

5.1 DC operating point and linear range (procedure step 1)

Table 2 lists the DC voltages measured with no input signal, in the format of the lab manual. Since R_B is the only DC path into each base, the base current follows from the base voltage and the collector current from the drop across R_C:

$$I_C = \frac{10 - V_C}{R_C}, \quad I_B = \frac{10 - V_B}{R_B}, \quad \beta = \frac{I_C}{I_B}, \quad g_m = \frac{I_C}{V_T}. \tag{3}$$

Table 2: Step 1 results. The linear range is derived below.

V_{C1}	V_{C2}	V_{B1}	V_{B2}	Linear range of output signal	Voltage offset
7.88 V	8.00 V	−2.33 V	−2.33 V	5.88 to 10 V per collector (±2.0 V about the Q-point)	−0.12 V ($V_{C1} - V_{C2}$)

Table 3: Bias parameters calculated from Table 2, the bias-matched simulation, and the ideal design (I_T = 9.3 V /6.8 kΩ split equally).

Parameter	Q_1 (bench)	Q_2 (bench)	Simulation (Q_1, Q_2)	Ideal design
V_C	7.88 V	8.00 V	7.89 V, 8.01 V	8.50 V
V_B	−2.33 V	−2.33 V	−2.32 V, −2.32 V	
I_C	0.964 mA	0.909 mA	0.960, 0.905 mA	0.684 mA
I_B	6.17 µA	6.17 µA		
β	156	147	156, 147	
g_m (V_T = 26 mV)	37.1 mS	35.0 mS		26.3 mS
$r_\pi = \beta/g_m$	4.22 kΩ	4.22 kΩ		
Tail current I_T	1.87 mA (1.885 mA with base currents)		1.878 mA	1.37 mA
Common emitters V_E	≈ −3.0 V (from $V_B - V_{BE}$, not recorded)		−2.98 V	

Tail current. The measured I_T = 1.87 mA is 37% above the ideal 1.37 mA. Two effects can be predicted. The mirror's output transistor runs at $V_{CE} \approx 7.0$ V against 0.67 V for the reference transistor, which raises its current by $(1 + 7.0/74)/(1 + 0.67/74) = 1.084$ for $V_A = 74$ V, and a finite $\beta \approx 150$ lowers it by 1.3%. Together they give 1.46 mA. The remaining factor of 1.28 corresponds to a V_{BE} mismatch of $V_T \ln 1.28 =$ 6.4 mV between the two mirror transistors, which is typical of unmatched, thermally separate 2N3904s. An R_K of about 5 kΩ instead of 6.8 kΩ would have the same effect; R_K was not measured.

Offset. Both bases are fed through 2 MΩ resistors with the same voltage drop, so both draw 6.17 µA and the collector currents split in the ratio of the β values. I_{C1} is 5.8% above I_{C2}, which gives $V_{C1} - V_{C2} = -0.12$ V. Referred to the input, the offset is 0.12 V /77.4 ≈ 1.5 mV.

Linear range of the output. Each collector can only move between 10 V (that transistor cut off) and $10 - I_T R_C = 5.88$ V (that transistor carrying all of I_T). Around the Q-points of 7.88 and 8.00 V this allows ±2.0 V per collector (4.1 Vpp) and ±4.12 V differential. Because $i_{C1} = (I_T/2)(1 + \tanh(v_d/2V_T))$, the gain falls gradually before these limits are reached; Table 4 gives the amplitudes for a given gain error.

Table 4: Linear range of the pair at the measured bias. The manual's 20 mVpp test input sits near the 1% row.

Gain error	Max v_d	Max output, one collector	Max output, differential
1%	18 mVpp	0.71 Vpp	1.42 Vpp
5%	42 mVpp	1.56 Vpp	3.13 Vpp
10%	61 mVpp	2.16 Vpp	4.33 Vpp

The simulation with ±150 mV at the base (Figure 7) drives the pair fully: V_{C1} swings from 5.90 to 9.99 V and the differential output from −4.09 to +4.07 V, within 0.05 V of the predicted limits. The one bench capture with a large input (Figure 8, 197 mVpp input) shows 3.8 Vpp at one collector, close to the 3.94

Vpp large-signal prediction and to the 4.12 Vpp limit. This capture is used only as a check of the output limit: this far into the nonlinear region, 3.8 V /0.197 V = 19.3 V/V is not a small-signal gain.

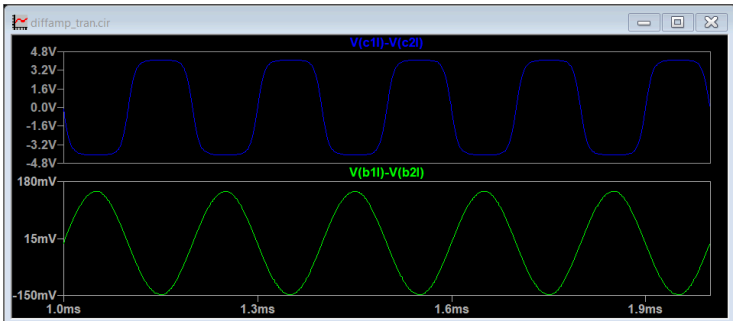


Figure 7: Simulated linear-range test: ±150 mV differential input (bottom) and differential output (top), which flattens at about ±4.08 V.

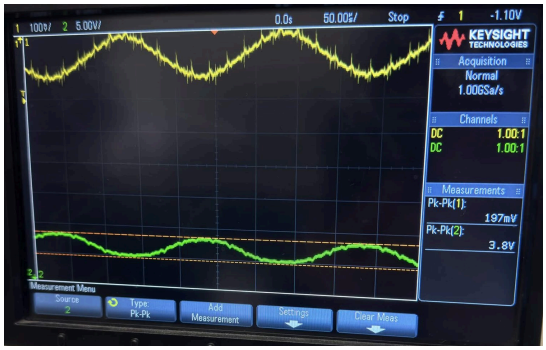


Figure 8: Bench, overdriven: CH1 (yellow, 100 mV/ div) input, 197 mVpp; CH2 (green, 5 V/div) one collector, 3.8 Vpp, inverted.

5.2 Signal source and input resistance (procedure step 2)

The function generator was set to a 5 kHz, 200 mVpp sine with the High Z output setting; the oscilloscope read 201 mVpp (Figure 9). R_S was then increased to bring v_d at the base of Q_1 down toward the 20 mVpp target. Figure 12 and Table 5 show the readings during this adjustment.

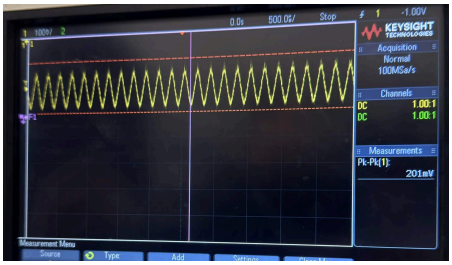


Figure 9: Generator output with High Z: 201 mVpp at 5 kHz.

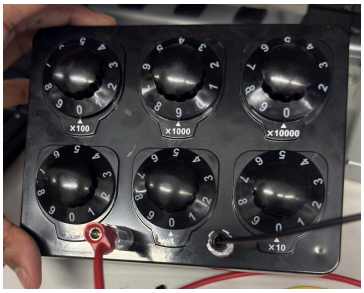


Figure 10: Decade box used for R_S, set to 9 kΩ.

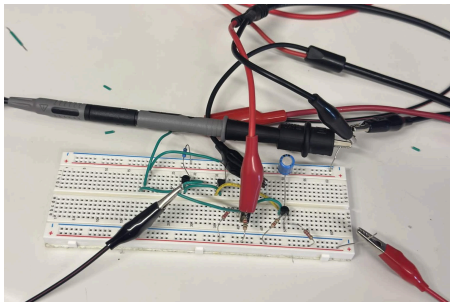


Figure 11: Circuit with R_S and the 10 µF coupling capacitor.



(a) (b) (c)

Figure 12: Adjusting R_S. CH1 (yellow) is the base of Q_1 and CH2 (green) is the source. (a) 201 mVpp at the base with a visible sine; (b) 68 mVpp at 100 mV/div and (c) 36 mVpp at 50 mV/div, both with no visible sine.

Table 5: Readings during the R_S adjustment. The lowest v_d reached was 36 mVpp.

Capture	v_d (CH1)	CH1 scale	v_s (CH2)	Sine visible
(a)	201 mVpp	100 mV/div	225 mVpp	Yes
(b)	68 mVpp	100 mV/div	237 mVpp	No
(c) $R_S = 829$ kΩ	36 mVpp	50 mV/div	235 mVpp	No

Using reading (c), with $k = v_d/v_s = 36/235 = 0.153$, the manual's divider equation rearranges to

$$R_p = R_{\text{id}} \parallel R_B = \frac{kR_S}{1-k} = \frac{0.153 \times 829\ \text{k}\Omega}{0.847} = 150\ \text{k}\Omega, \quad R_{\text{id}} = \frac{R_p R_B}{R_B - R_p} = 162\ \text{k}\Omega. \tag{4}$$

The expected input resistance with Q_2's base grounded is

$$R_{\text{id}} = 2r_\pi = \frac{2V_T}{I_B} = \frac{2 \times 26\ \text{mV}}{6.17\ \mu\text{A}} = 8.43\ \text{k}\Omega, \tag{5}$$

and the simulation gives 8.64 kΩ. At $R_S = 829$ kΩ the true v_d was therefore only about 2.4 mVpp (Figure 13), far below the 36 mVpp noise floor of the oscilloscope in Normal acquisition. Captures (b) and (c) show no sine, and the reading roughly halved with the vertical scale. The 36 mVpp reading is noise, the true k is smaller than 0.153, and the bench result is only an upper bound: $R_{\text{id}} \leq 162$ kΩ. The 20 mVpp target needed $R_S \approx 77$ kΩ, which the decade box alone can provide (its ×10000 dial reaches about 100 kΩ), and Averaging acquisition (manual step 3f) would have lowered the noise floor.

Two further observations. The source read 225 to 237 mVpp although a High Z generator set to 200 mVpp cannot deliver more, because the peak-to-peak measurement includes the noise spikes. Capture (a), with a clean 201 mVpp at the base, was taken at a small, unrecorded R_S; with $R_{\text{id}} \approx 8.6$ kΩ it implies $R_S \approx 1$ kΩ.

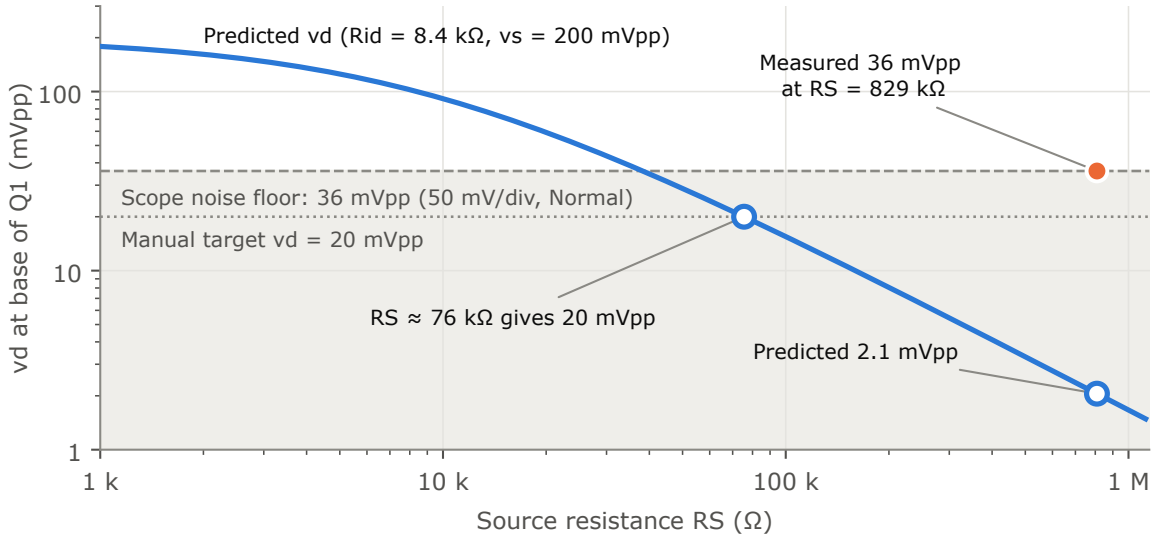


Figure 13: Predicted $v_d = v_s R_p/(R_S + R_p)$ for $v_s = 200$ mVpp and $R_{\text{id}} = 8.4$ kΩ. The 20 mVpp target needs $R_S \approx 76$ kΩ; at 829 kΩ the signal (≈ 2 mVpp) is far below the 36 mVpp noise floor, which is what was measured.

5.3 Differential gain (procedure step 3)

Bench. No valid small-signal gain was recorded. The input values noted for this step ($v_d = 36$ mVpp, $v_s = 235$ mVpp) come from the $R_S = 829$ kΩ setting, where v_d is noise, and the output value (3.8 Vpp) comes from the overdriven capture of Figure 8. Their ratios (106 and 16.2 V/V) do not describe one operating condition and are not reported as results. A correct measurement needs $R_S \approx 77$ kΩ ($v_d \approx 20$ mVpp) and the differential output measured as CH1 − CH2.

Expected gain. For matched transistors at the measured bias,

$$g_m = \frac{I_{C1} + I_{C2}}{2V_T} = \frac{0.936\ \text{mA}}{26\ \text{mV}} = 36.0\ \text{mS}, \quad A_d = \frac{v_{o1} - v_{o2}}{v_d} = -g_m R_C = -79.2\ \text{V/V}, \tag{6}$$

or −39.6 V/V at each collector (−57.9 V/V at the ideal 1.37 mA bias). Including the transistors' output resistance, $r_o = (V_A + V_{CE})/I_C = (74 + 10.9)\ \text{V}\ /0.936\ \text{mA} = 90.7$ kΩ, in parallel with R_C gives $A_d = -77.4$ V/V.

Simulation. With $v_s = 200$ mVpp and $R_S = 76$ kΩ, the AC analysis gives $A_d = -77.4$ V/V, -38.7 V/V at each collector, and $A_{\text{ds}} = -7.86$ V/V, with the output inverted (180°). The transient analysis (Figure 14) gives $v_d = 20.6$ mVpp, $v_{\text{o1}} = 0.79$ Vpp and $v_{\text{o1}} - v_{\text{o2}} = 1.57$ Vpp, a gain of 76.4 V/V. The 1.2% lower value is the compression expected at this input level (Table 4, 1% row).

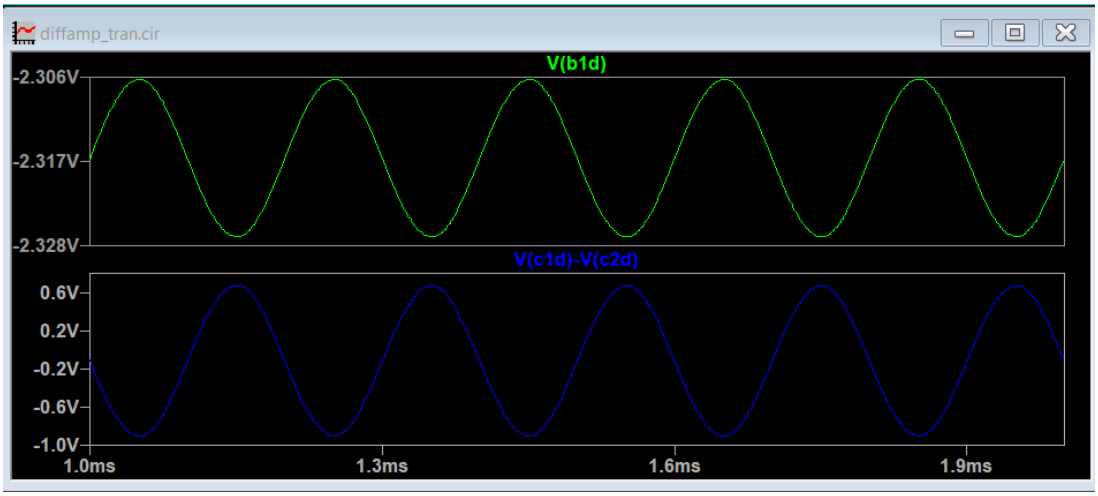


Figure 14: Simulated differential mode, $R_S = 76$ kΩ: base of Q_1 (top, 20.6 mVpp around -2.317 V) and differential output (bottom, 1.57 Vpp, inverted).

Table 6: Step 3 results in the manual's format. No valid bench measurement was obtained; the values are from the bias-matched simulation ($v_s = 200$ mVpp, $R_S = 76$ kΩ, differential output).

V_o	A_d	A_{ds}	Expected voltage gain
1.57 Vpp	-77.4 V/V	-7.86 V/V	$-g_m R_C = -79.2$ V/V (-77.4 V/V including r_o)

5.4 Common-mode gain and CMRR (procedure step 4)

Bench. With Q_2's base moved to B and R_S still at 829 kΩ, the output kept clipping. The captured waveform (Figure 15) is 520 mVpp at one collector, but its period is about 2.4 div at 50 µs/div (≈ 8.4 kHz) rather than 200 µs, and the collector sat at about 9.0 V DC (the CH2 trigger level, 9.07 V, lies on the trace) instead of 7.9 to 8.0 V, that is about 0.48 mA instead of 0.94 mA. The waveform is therefore not the response to the 5 kHz input, and the operating point had shifted. The base signal was not recorded (CH1: No signal). For comparison, the common-mode input resistance, $(R_B/2) \parallel (\beta+1)R_o \approx 0.87$ MΩ, is comparable to R_S, so the bases would have seen about 120 mVpp and the expected single-ended output was about 3 mVpp. A_{cm} and the CMRR were therefore not measured.

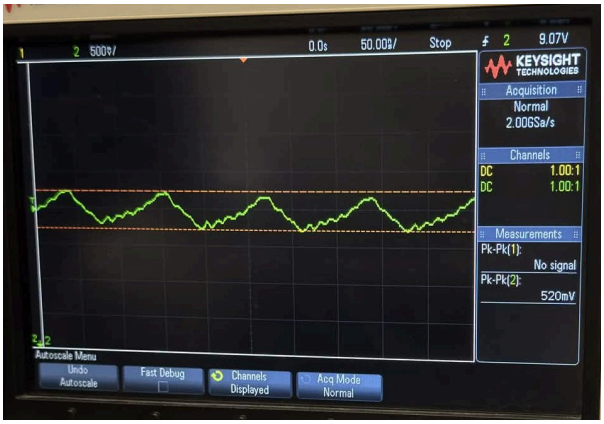


Figure 15: Bench, common mode: one collector (CH2, DC coupling, 500 mV/div), 520 mVpp, distorted, period ≈ 120 µs, DC level ≈ 9.0 V.

Expected values. The mirror's output resistance is $R_o = (V_A + V_{CE})/I_T = (74 + 7.0)$ V /1.87 mA = 43 kΩ. For one collector,

$$A_{cm} \approx \frac{R_C}{2R_o} = \frac{2.2\ \text{k}\Omega}{86.5\ \text{k}\Omega} = 0.025\ \text{V/V}, \quad \text{CMRR} = 20\log\frac{|A_{d,se}|}{|A_{cm}|} = 63.6\ \text{dB}. \tag{7}$$

For the differential output with perfectly matched halves $A_{cm} = 0$ and the CMRR is infinite. The measured 5.8% g_m mismatch gives $A_{cm,d} \approx 0.058 \times R_C/(2R_o) = 0.0015$ V/V and a CMRR of about 94 dB.

Simulation. The AC analysis gives $A_{cm} = 0.0261$ V/V at one collector and 0.00144 V/V differential, so the CMRR is 63.4 dB single-ended and 94.6 dB differential, within 0.3 dB of the hand estimates.

Table 7: Step 4 results in the manual's format, from the bias-matched simulation. The bench common-mode measurement was not valid.

v_o	A_{cm}	CMRR (dB)
5.2 mVpp (one collector, $v_{cm} = 200$ mVpp)	0.0261 V/V (one collector) 0.00144 V/V (differential)	63.4 (single-ended) 94.6 (differential)

6 Comparison of results

Table 8: Hand analysis, simulation and bench results.

Quantity	Hand analysis	Simulation	Bench	Assessment
Mirror i_{copy}/i_{ref}	1	1.000 (ideal model)	n/a	Agrees
Mirror R_x bound	7.16 kΩ	7.13 kΩ	n/a	Within 0.5%
Tail current I_T	1.37 mA (1.46 with V_A, β)	1.878 mA (matched)	1.873 mA	37% above design
V_{C1}, V_{C2}	8.50 V (ideal)	7.89, 8.01 V	7.88, 8.00 V	Follows I_T
Offset $V_{C1} - V_{C2}$	0	−0.120 V	−0.12 V	β mismatch
Output limits, one collector	5.88 to 10 V	5.90 to 9.99 V	3.8 Vpp (overdriven)	Agrees
R_{id}	8.43 kΩ	8.64 kΩ	≤ 162 kΩ	Bench: bound only
A_d	−79.2 V/V (−77.4 with r_o)	−77.4 V/V	not measured	Hand and sim agree
A_{ds} at $R_S = 76$ kΩ	−7.70 V/V	−7.86 V/V	not measured	Within 3%
A_{cm}, one collector	0.025 V/V	0.026 V/V	not measured	Within 3%
CMRR, single-ended	63.6 dB	63.4 dB	not measured	Agrees
CMRR, differential	∞ ideal, ≈ 94 dB with mismatch	94.6 dB	not measured	Agrees

7 Discussion

- **Bias.** The 37% excess tail current is not a fault of the circuit. The Early effect accounts for 8%, and the rest corresponds to a 6 mV V_{BE} mismatch between the discrete mirror transistors, which an integrated mirror (matched, thermally coupled devices) would avoid. The higher current raises g_m, so the expected A_d rises from −57.9 to −79.2 V/V.
- **Offset.** With current-fed bases, the 6% β mismatch (156 against 147) sets a 6% collector-current mismatch and the −0.12 V output offset (about 1.5 mV at the input). The manual's advice to choose transistors with close β addresses exactly this.

- **Why the bench gain and CMRR failed.** In step 2, R_S was about ten times too large (829 kΩ against 77 kΩ), so v_d fell below the noise floor of Normal acquisition. In step 3 the recorded input and output came from different settings, and the only output capture was overdriven. In step 4 the output was not a response to the input and the bias had shifted. Repeating steps 2 to 4 with $R_S \approx 77$ kΩ from the decade box, Averaging acquisition, the differential output measured with CH1 − CH2, and in common mode a small v_s with both bases recorded would give valid measurements.
- **Simulation.** Matched to the measured bias, the simulation reproduces every hand estimate within 3%. The 2% gap between $-g_m R_C$ and the simulated A_d is the transistors' output resistance, and including r_o closes it. The pre-lab simulations show that a SPICE model's base-current terms (I_{SE} as well as β_F) must be removed for a mirror to copy its reference exactly.

8 Conclusion

The current mirror and the differential pair behave as analysed. In simulation the ideal mirror copied its reference exactly and stayed active up to $R_x = 7.13$ kΩ, against 7.16 kΩ by hand; the 2N3394 model's base-current terms explain the lower copies obtained with the default model. On the bench the mirror delivered 1.87 mA, 37% above design, consistent with the Early effect and a 6 mV mismatch between the discrete mirror transistors. The DC operating point, the −0.12 V offset and the output limits of 5.88 to 10 V per collector were measured or confirmed. The input resistance, differential gain and CMRR could not be measured reliably because of the source resistance, the oscilloscope noise floor and an overdriven or unstable output. From analysis and the bias-matched simulation they are $R_{\mathrm{id}} \approx 8.6$ kΩ, $A_d \approx -77$ V/V and a CMRR of about 63 dB single-ended (95 dB differential).

Appendix A: Bench notes

Handwritten notes taken during the lab, used for Table 2 and Table 5.

Figure 7: Internal circuit of a differential amplifier

V_{C1}	V_{C2}	V_{B1}	V_{B2}	Linear range of output signal	Voltage offset
7.88 V	8.00 V	-2.3 V	-2.8 V		0.12

Figure 16: Step 1: V_{C1}, V_{C2}, V_{B1}, V_{B2} and the 0.12 V offset.

Table 4: Enabling High Z and finalizing the settings

Adjust the variable resistor R_s until the amplitude of v_d is 10 mV (20 mV_{pp}). Given that

$$\frac{v_d}{v_s} = \frac{R_{id} \parallel R_B}{R_s + (R_{id} \parallel R_B)},$$

compute R_{id}.

R_{id} = _____ $R_S = 9k + 820k$

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Figure 17: Step 2: final source resistance, $R_S = 9\ \text{k}\Omega + 820\ \text{k}\Omega$.

Figure 8: Sine wave, 200 mV_{pp} at 5 kHz.

h. You can now add measurements related to this signal.

Compute: 36 mV 235 mV

$$A_d = \frac{v_o}{v_d}, \qquad A_{ds} = \frac{v_o}{v_s}.$$

Be sure to properly measure the differential output $v_{o1} - v_{o2}$. Assuming matched transistors, what differential gain do you expect?

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V_o	A_d	A_{ds}	**Expected voltage gain**
3.8V			

4) Disconnect the base of Q_2 from ground and connect it to point B (Figure 7) to enter *common-mode*. Observe the output v_o and measure its amplitude. If the output is clipped or distorted, reduce the amplitude of v_s. Determine the common-mode gain $A_{cm} = v_o/v_{cm}$ and compute the CMRR in dB:

$$\text{CMRR} = 20 \log\left(\frac{A_d}{A_{cm}}\right).$$

v_o	A_{cm}	CMRR (dB)

Figure 18: Step 3: $v_d = 36$ mV, $v_s = 235$ mV and $V_o = 3.8$ V (see Section 5.3 for why these are not used as a gain).

Appendix B: LTspice netlists

Current mirror sweep (sim/mirror_sweep.cir)

```
* BJT current mirror (recreation of the Multisim circuit, Lab 2 pre-lab)
*
* Rx is swept with .dc instead of .step + .op. LTspice evaluates .meas
* separately for every .step of an .op, so AT= and WHEN cannot work across
* Rx values. Here Rx is a behavioural resistor whose value is V(rx) in kOhm,
* so the x-axis (Vrx) reads directly in kOhm and every .meas line works.
*
* Three model cases are stepped with cfg:
*   cfg=1  BF=121.3, ISE on   (model as given)          expect Icopy(1k) ~
1.337 mA
*   cfg=2  BF=1e6,   ISE on   (BF raised only)          expect Icopy(1k) ~
1.359 mA (Multisim screenshot)
*   cfg=3  BF=1e6,   ISE=0    (ideal, no base current)  expect Icopy(1k) = Iref
~ 1.380 mA
* The ISE (B-E leakage) term adds about 10 uA of base current at 1.4 mA,
* so raising BF alone only lifts the effective beta to about 130.

R1   0   B   6.8k
* BRx behaves exactly like a resistor of V(rx) kOhm between node 0 and VC
BRx  0   VC  I=-V(VC)/(V(rx)*1k)
Q1   B   B   VE  QN
Q2   VC  B   VE  QN
V1   VE  0   -10
* Rx value in kOhm (swept by .dc)
Vrx  rx  0   1

.param bfv={if(cfg==1,121.3,1e6)}
.param isev={if(cfg==3,0,1.222e-13)}
.model QN NPN(IS=6.101e-14 BF={bfv} NF=1 VAF=2e6 IKF=0.09779 ISE={isev}
NE=1.305 BR=0.2359 NR=1)

.dc Vrx 1 10 0.01
.step param cfg 1 3 1

* Results: View > SPICE Error Log, one row per cfg. AT values are Rx in kOhm.
.meas DC Iref       FIND I(R1)  AT=1
.meas DC Icopy_1k   FIND I(BRx) AT=1
.meas DC ratio_1k   PARAM Icopy_1k/Iref
.meas DC Icopy_4k   FIND I(BRx) AT=4
.meas DC Icopy_7k   FIND I(BRx) AT=7
.meas DC Icopy_7k2  FIND I(BRx) AT=7.2
.meas DC Icopy_8k   FIND I(BRx) AT=8
.meas DC Icopy_10k  FIND I(BRx) AT=10
.meas DC VCE2_7k2   FIND V(VC)-V(VE) AT=7.2
.meas DC VCE2_8k    FIND V(VC)-V(VE) AT=8
* Upper bound of Rx: Q2 reaches VCE,sat = 0.2 V
.meas DC Rx_knee    WHEN V(VC)-V(VE)=0.2
* Rx where Icopy has dropped 1% below its flat value
.meas DC Rx_1pct    WHEN I(BRx)=0.99*Icopy_1k
.end
```

Differential amplifier, AC analysis (sim/diffamp_ac.cir)

```
* ELG3136 Lab 2: differential amplifier (Figure 7), small-signal values at 5
kHz
*   copy d: differential mode, RS = 76k, Q2 base AC-grounded
*   copy c: common mode, Q2's capacitor moved to B
* Results go to View > SPICE Error Log, printed as (gain in dB, phase in
degrees).
* Devices match the bench: beta ~156 / ~147, tail current ~1.87 mA
* (Q4 area 1.28 stands for the ~6 mV VBE mismatch of the discrete mirror).
* Note: 2Meg, not 2M (2M means 2 milliohm in SPICE).

VCC  vcc 0 10
VEE  vee 0 -10
VS   src 0 AC 1

* ---------- copy d: differential mode ----------
RSd  src  xd   76k
C1d  xd   b1d  10u
RB1d vcc  b1d  2Meg
RB2d vcc  b2d  2Meg
C2d  b2d  0    10u
RC1d vcc  c1d  2.2k
RC2d vcc  c2d  2.2k
Q1d  c1d  b1d  ed  QP1
Q2d  c2d  b2d  ed  QP2
RKd  0    kd   6.8k
Q3d  kd   kd   vee QM
Q4d  ed   kd   vee QM 1.28

* ---------- copy c: common mode ----------
C1c  src  b1c  10u
RB1c vcc  b1c  2Meg
RB2c vcc  b2c  2Meg
C2c  b2c  b1c  10u
RC1c vcc  c1c  2.2k
RC2c vcc  c2c  2.2k
Q1c  c1c  b1c  ec  QP1
Q2c  c2c  b2c  ec  QP2
RKc  0    kc   6.8k
Q3c  kc   kc   vee QM
Q4c  ec   kc   vee QM 1.28

.model QP1 NPN(IS=6.734e-15 BF=137 VAF=74 CJE=8p CJC=4p TF=350p)
.model QP2 NPN(IS=6.734e-15 BF=129 VAF=74 CJE=8p CJC=4p TF=350p)
.model QM  NPN(IS=6.734e-15 BF=150 VAF=74 CJE=8p CJC=4p TF=350p)

.ac lin 3 4k 6k

* vd/vs (for Rid)
.meas AC vd_vs   FIND V(b1d)/V(src)            AT=5k
* differential mode
.meas AC Ad      FIND (V(c1d)-V(c2d))/V(b1d)   AT=5k
.meas AC Ad_se   FIND V(c1d)/V(b1d)            AT=5k
.meas AC Ads     FIND (V(c1d)-V(c2d))/V(src)   AT=5k
* common mode
.meas AC Acm_se  FIND V(c1c)/V(b1c)            AT=5k
.meas AC Acm_d   FIND (V(c1c)-V(c2c))/V(b1c)   AT=5k
* CMRR (the dB value printed is the CMRR)
.meas AC CMRR_se PARAM Ad_se/Acm_se
.meas AC CMRR_d  PARAM Ad/Acm_d
.end
```

Differential amplifier, transient analysis (sim/diffamp_tran.cir)

```
* ELG3136 Lab 2: differential amplifier (Figure 7), DC point, waveforms, linear
range
*   copy d: differential mode, 200 mVpp source, RS = 76k  (vd ~ 20 mVpp)
*   copy l: 300 mVpp straight into the base               (linear range)
* Numbers go to View > SPICE Error Log.
* Note: 2Meg, not 2M (2M means 2 milliohm in SPICE).

VCC  vcc 0 10
VEE  vee 0 -10
VS   src 0 SINE(0 0.1 5k)
VL   sl  0 SINE(0 0.15 5k)

* ---------- copy d: differential mode ----------
RSd  src  xd   76k
C1d  xd   b1d  10u
RB1d vcc  b1d  2Meg
RB2d vcc  b2d  2Meg
C2d  b2d  0    10u
RC1d vcc  c1d  2.2k
RC2d vcc  c2d  2.2k
Q1d  c1d  b1d  ed  QP1
Q2d  c2d  b2d  ed  QP2
RKd  0    kd   6.8k
Q3d  kd   kd   vee QM
Q4d  ed   kd   vee QM 1.28

* ---------- copy l: linear range ----------
C1l  sl   b1l  10u
RB1l vcc  b1l  2Meg
RB2l vcc  b2l  2Meg
C2l  b2l  0    10u
RC1l vcc  c1l  2.2k
RC2l vcc  c2l  2.2k
Q1l  c1l  b1l  el  QP1
Q2l  c2l  b2l  el  QP2
RKl  0    kl   6.8k
Q3l  kl   kl   vee QM
Q4l  el   kl   vee QM 1.28

.model QP1 NPN(IS=6.734e-15 BF=137 VAF=74 CJE=8p CJC=4p TF=350p)
.model QP2 NPN(IS=6.734e-15 BF=129 VAF=74 CJE=8p CJC=4p TF=350p)
.model QM  NPN(IS=6.734e-15 BF=150 VAF=74 CJE=8p CJC=4p TF=350p)

.tran 0 3m 0 0.2u
.options plotwinsize=0

* ---------- DC operating point (t = 0) ----------
.meas TRAN VC1   FIND V(c1d)  AT=0
.meas TRAN VC2   FIND V(c2d)  AT=0
.meas TRAN VB1   FIND V(b1d)  AT=0
.meas TRAN VB2   FIND V(b2d)  AT=0
.meas TRAN VE    FIND V(ed)   AT=0
.meas TRAN IT    FIND Ic(Q4d) AT=0

* ---------- differential mode amplitudes ----------
.meas TRAN vs_pp   PP V(src)          FROM=1m TO=3m
.meas TRAN vd_pp   PP V(b1d)          FROM=1m TO=3m
.meas TRAN vo1_pp  PP V(c1d)          FROM=1m TO=3m
.meas TRAN vod_pp  PP V(c1d)-V(c2d)   FROM=1m TO=3m

* ---------- linear range ----------
.meas TRAN vc1_max MAX V(c1l)          FROM=1m TO=3m
.meas TRAN vc1_min MIN V(c1l)          FROM=1m TO=3m
.meas TRAN vod_max MAX V(c1l)-V(c2l)   FROM=1m TO=3m
.meas TRAN vod_min MIN V(c1l)-V(c2l)   FROM=1m TO=3m
.end
```